Manea Abdullah Al-Awbathani

SENIOR SOFTWARE ENGINEER | TECHNICAL LEAD | SOFTWARE ARCHITECT

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PROFESSIONAL SUMMARY

Hands-on senior software engineer and technical lead who designs, builds, and ships production-ready software end to end. Experience across multi-tenant SaaS, real-time telehealth, energy monitoring, compliance, marketplaces, and AI-enabled industrial systems. Strong in system architecture, full-stack engineering, backend APIs, cloud infrastructure, real-time data, security, and AI integration, with experience taking products from requirements and architecture through implementation, deployment, documentation, and technical leadership.

CORE SKILLS

Software Engineering: TypeScript, Python, JavaScript, PHP, Node.js, NestJS, Express.js, FastAPI

Frontend: React, Next.js, Vue.js, Quasar, Tailwind CSS, Zustand, TanStack Query, React Hook Form

Backend and APIs: REST, GraphQL, Swagger/OpenAPI, WebSockets, Socket.io, event-driven architecture

Architecture: System design, multi-tenant SaaS, real-time systems, workflow systems, RBAC, API architecture

AI and Machine Learning: LLMs, OpenAI GPT, Claude, LLaMA, agentic AI, RAG, prompt engineering, vector databases, MLOps

Data and Cloud: PostgreSQL, MySQL, MongoDB, Redis, TypeORM, Prisma, AWS EC2/S3/RDS/Lambda, Docker, CI/CD

Security and Quality: JWT, Passport.js, bcrypt, Helmet, rate limiting, OWASP practices, Jest, Supertest, Testing Library

PROFESSIONAL EXPERIENCE

Software Engineering Lead - Aip

Jan 2026 - Present | Riyadh, Saudi Arabia | Full-time, On-site

- Lead software architecture and engineering initiatives across enterprise and industrial systems.
- Define cloud architecture standards and engineering practices from data pipelines through production deployment.
- Architect AI-enabled solutions using LLMs, agentic AI, and RAG across ERP, MES, CMMS, and SCADA environments.
- Help build the internal engineering team and translate business priorities into production-ready systems.

Lead Software Architect - ISOPluss

Apr 2025 - Present | Riyadh, Saudi Arabia | Concurrent, Part-time

- Co-founded and architected a multi-tenant B2B SaaS platform for ISO 22000 and HACCP compliance.
- Designed the NestJS, Next.js, PostgreSQL, Redis, and TypeORM architecture with JWT/RBAC security and documented APIs.
- Built a schema-driven engine for digital workflows, document control, audit management, corrective actions, and analytics.
- Own cloud infrastructure, CI/CD, DevOps automation, and production data-security practices end to end.

Senior Software Engineer - Me'Kaaz

Jul 2025 - Jan 2026 | Riyadh, Saudi Arabia | Contract, On-site

- Architected and delivered a BLE-smartwatch telehealth platform connecting patient applications with hospital systems.
- Engineered real-time cloud infrastructure achieving under 3-second vitals latency and 99.9% uptime.
- Delivered hospital dashboards, ward heatmaps, automated clinical alerts, and rapid branded deployment workflows.
- Led cross-functional design, engineering, and QA coordination through regional rollout, knowledge transfer, and mentoring.

Software Engineering Lead - MYCES SDN BHD

Jul 2024 - Dec 2024 | Malaysia | Full-time, Hybrid

- Delivered enterprise systems across agriculture, dam safety, and energy sectors.
- Shipped three major platforms from requirements through MVP using Agile practices and reusable components.
- Reduced delivery timelines by 20% through reusable architecture and component practices.
- Authored SRS, SDD, and STR/STD documentation and mentored junior developers.

Software Engineer - MYCES SDN BHD

Dec 2023 - Aug 2024 | Malaysia | Full-time, On-site

- Built upgrades and features for EMARS, an energy-monitoring SaaS serving 100+ enterprise facilities.
- Delivered threshold-based alerting and treemap analytics that contributed to a reported 17% reduction in energy waste across client facilities.
- Improved user engagement by 22% through responsive web and mobile interfaces and strengthened security with RBAC.
- Started with the company as a Full-Stack Intern (Oct-Nov 2023).

SELECTED ENGINEERING WORK

ISOPluss FSMS - Food Safety Management System

ISOPluss | Riyadh, Saudi Arabia

- Architected a modular enterprise SaaS for ISO 22000 and HACCP compliance.
- Designed a multi-tenant workflow platform covering document control, audits, objectives tracking, corrective actions, and analytics.
- Built the platform around NestJS, Next.js, PostgreSQL, Redis, TypeORM, JWT/RBAC security, and a documented API.

Aip - Software and AI Platform Architecture

Aip | Riyadh, Saudi Arabia

- Architected software and AI solutions spanning ERP, MES, CMMS, and SCADA environments.
- Established cloud architecture and engineering practices for production deployment, data pipelines, and AI integration.
- Applied LLM, agentic AI, and RAG patterns to enterprise and industrial use cases.

HealthWatch - Real-Time Telehealth Platform

Me'Kaaz | Riyadh, Saudi Arabia

- Engineered a real-time ecosystem connecting BLE smartwatches, patient applications, and a multi-tenant hospital portal.
- Delivered automated clinical alerts, ward heatmaps, and role-based doctor-patient workflows with under 3-second vitals latency.
- Supported regional rollout with NCA/MOH compliance requirements and hospital deployment workflows.

GavelMarket - Real-Time Online Auction Platform

Part-time | Oct 2025 - Present | Saudi Arabia, Remote

- Architected and built the full stack for high-concurrency online auctions using Express.js, Prisma, PostgreSQL, Socket.io, JWT, Cloudinary, SendGrid, Twilio, Next.js, and React.
- Implemented timer-based lifecycle events, rate limiting, structured logging, and data-integrity controls for secure payment flows.

COVITA - Multi-Vendor Coffee Marketplace

Contract | Feb 2026 - Present | Riyadh, Saudi Arabia, Remote

- Led end-to-end architecture and development for a B2B/B2C coffee marketplace.
- Built vendor onboarding, product catalogs, order management, payment integration, and mobile/web applications.
- Designed the platform for multiple business models across wholesale, retail, and subscription commerce.

MYCES Engineering Portfolio - AgroFarm, EMARS, Dam Monitoring, EMS

MYCES SDN BHD | Malaysia

- AgroFarm: built a farm-monitoring platform covering real-time data, resource planning, automated pricing tools, and reporting; won the Gold Medal at the Johor Agriculture Department Innovation Competition 2024.
- EMARS: built energy-monitoring upgrades, threshold alerts, and treemap analytics for 100+ enterprise facilities; reported 17% reduction in energy waste and 22% higher user engagement.
- Dam Monitoring and EMS: delivered enterprise monitoring platforms from requirements through MVP using reusable architecture, Agile practices, and formal SRS/SDD/STR/STD documentation.

EDUCATION

Bachelor of Computer Science (Software Engineering) - University Malaysia Pahang, Pahang, Malaysia
Oct 2020 - Mar 2024

ACCREDITATION, CERTIFICATIONS AND AWARDS

- Saudi Council of Engineers - Professional Accreditation Certificate, Specialist (Computer Science)
- IBM - Exploratory Data Analysis for Machine Learning
- Gold Medal - Johor Agriculture Department Innovation Competition 2024 (AgroFarm, MYCES)
- Additional coursework: Quantum Computing (Saint Petersburg State University); Blockchain Basics (University at Buffalo)

PUBLICATION

IEEE - Charity and Donation Tracking System Using Queue Structure (published research paper).